

03_PC_to_Router_Default_Gateway – Connect PCs Through a Router and Configure Default Gateway

Packet Tracer – PC to Router Connection

Objectives

Part 1: Build a Network with a Router

Part 2: Configure Static IP Addressing on PCs and Router Interfaces

Part 3: Configure Default Gateway on PCs

Part 4: Verify Connectivity Using Ping

Background / Scenario

You are learning how devices communicate across subnets using a router. In this lab, you will:

- Connect two PCs to a router using a switch
- Assign static IPv4 addresses to PCs
- Configure the router with appropriate interface IPs
- Set the default gateway on each PC
- Test connectivity across the network

This lab reinforces understanding of:

- Role of routers in inter-network communication
- Default gateway configuration
- Static IP addressing
- ICMP (ping) for testing connectivity

Instructions

Part 1: Connect the Devices

Step 1: Add Devices

- a. Open **Cisco Packet Tracer**.
- b. From **End Devices**, drag **PC0** and **PC1** onto the workspace.

- c. From **Network Devices** → **Routers**, drag **Router0** onto the workspace.
 - d. From **Network Devices** → **Switches**, drag **Switch0** onto the workspace.
 - e. Rename the PCs:
 - PC0 → **PC-A**
 - PC1 → **PC-B**
-

Step 2: Connect the Devices

- a. Click **Connections** (lightning bolt icon).
- b. Use **Copper Straight-Through cable** to connect:
 - PC-A → Switch0 (FastEthernet0)
 - PC-B → Switch0 (FastEthernet0)
- c. Connect Router0 to Switch0 using **Copper Straight-Through cable**:
 - Router0 → Switch0 (use GigabitEthernet0/0)
- d. Wait for link lights to turn **green**.

Your physical topology is now complete.

Part 2: Configure Router Interfaces

Step 1: Configure Router0 Interface

- a. Click **Router0** → **CLI** tab.
- b. Enter the following commands:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.254 255.255.255.0
no shutdown
exit
```

Explanation: Router interface G0/0 acts as the **default gateway** for the PCs in the 192.168.1.0/24 network.

Part 3: Configure Static IP Addressing on PCs

Step 1: Configure PC-A

a. Click **PC-A → Desktop → IP Configuration**

b. Enter:

- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.254

Step 2: Configure PC-B

a. Click **PC-B → Desktop → IP Configuration**

b. Enter:

- IP Address: 192.168.1.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.254
-

Part 4: Test Connectivity

Step 1: Ping Router from PC-A

a. Click **PC-A → Desktop → Command Prompt**

b. Type:

```
ping 192.168.1.254
```

Expected Result:

```
Reply from 192.168.1.254: bytes=32 time<1ms TTL=128
```

Step 2: Ping PC-B from PC-A

a. Type:

```
ping 192.168.1.2
```

Expected Result:

```
4 successful replies.
```

Step 3: Ping PC-A from PC-B

a. Repeat the ping command in **PC-B's** command prompt:

ping 192.168.1.1

You should again see 4 successful replies.

Troubleshooting Tips

- ✓ Ensure all cables are **Straight-Through** (not cross-over for switch connections).
 - ✓ Confirm IP addresses and subnet masks match the network.
 - ✓ Verify default gateway is set correctly on each PC.
 - ✓ Check that router interfaces are **no shutdown** and link lights are green.
-

What You Learned

- How to connect devices through a router
 - How static IP addressing and default gateways work
 - Role of routers in inter-network communication
 - Using ICMP (ping) to verify connectivity
-

Challenge Extensions (Optional)

1. Add another subnet (192.168.2.0/24) connected to Router0 → test inter-subnet communication.
 2. Configure DHCP on the router to automatically assign IP addresses to PCs.
 3. Add a second router and connect different subnets → test routing between multiple networks.
-

You have now completed a foundational networking lab: **using a router as a default gateway to enable PC-to-PC communication across subnets**. This builds the foundation for routing, switching, and CCNA-level network configuration.